

Handout 5.3: Fiber Test Station Record

Name: _____ Date: _____ Period: _____ Station: _____

Directions: Fill in Part 1 during the slides. Take this sheet to the station for Part 2. Fill in Part 3 during the teacher demonstration. Finish Part 4 after the envelope is opened.

Station rules: one drop of water, not a puddle. Swatches stay at the station. Wet swatches go on the paper towel with the letter showing. Nothing goes in a mouth.

Word bank in other languages

[Check with a native speaker before printing.]

English	Turkish	Portuguese	Spanish
fiber	lif	fibra	fibra
natural	doğal	natural	natural
manufactured	yapay (sentetik)	manufaturada (sintética)	manufacturada (sintética)
cotton	pamuk	algodão	algodón
wool	yün	lã	lana
silk	ipek	seda	seda
linen	keten	linho	lino
polyester	polyester	poliéster	poliéster
nylon	naylon	náilon	naílon
rayon	rayon (viskon)	raiom (viscose)	rayón (viscosa)
spandex	likra (elastan)	elastano (lycra)	elastano (licra)
absorb	emmek	absorver	absorber
stretch	uzamak (esnemek)	esticar	estirar
wrinkle	kırışmak	amassar (vincar)	arrugarse
warm	sıcak tutan	quente	cálido
melt	erimek	derreter	derretirse

Part 1: Two families of fiber

1a. Sort the eight fibers. cotton, wool, silk, linen, polyester, nylon, rayon, spandex

Natural (from a plant or an animal)	Manufactured (made in a factory)

1b. Name the source of each natural fiber.

Fiber	Comes from
Cotton	
Wool	
Silk	
Linen	

1c. The class table. Fill in as the slides go. Use words like high, low, some, none, cheap, costly.

Fiber	Warmth	Absorbs or wicks water	Wrinkles	Stretch	Cost and care
Cotton					
Wool					
Silk					
Linen					
Polyester					
Nylon					
Rayon					
Spandex					

Part 2: The station record

Three tests on six swatches. Circle or write in every box. Ninety seconds per swatch.

Test 1, feel: rub it between finger and thumb. Circle one in each pair. **Test 2, stretch:** pull gently across, then let go. Did it come back? **Test 3, water drop:** one drop, count to ten. Soaked in, beaded up, or spread out?

Swatch	Feel: smooth	Feel: warm or	Feel: thick or	Stretch: came	Water: soaked	My guess at the	Good for making
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	or rough	cool	thin	back / stayed stretche d / no stretch	in / beaded / spread	fiber	
A	smooth / rough	warm / cool	thick / thin				
B	smooth / rough	warm / cool	thick / thin				
C	smooth / rough	warm / cool	thick / thin				
D	smooth / rough	warm / cool	thick / thin				
E	smooth / rough	warm / cool	thick / thin				
F	smooth / rough	warm / cool	thick / thin				

Part 3: The burn test (teacher demonstration only)

You are seated behind the line. Nobody stands up. Nobody films it.

Fiber	What the flame did (burned fast / burned slow / melted)	The smell, in one word	What was left (soft ash / crushable bead / hard bead)
Cotton			
Wool			
Polyester			

Finish the sentence: A fiber that melts is dangerous near heat because



If there is no flame today, do the wrinkle test instead. Crush each swatch in your fist for ten seconds, then open your hand. Rate the wrinkle 1 (none), 2 (some), 3 (a lot).

Swatch	A	B	C	D	E	F
Wrinkle 1 to 3						

Part 4: The reveal and the four use questions

4a. Write the real fiber next to each letter, then check your guess.

Swatch	Real fiber	Was my guess right?
A		yes / no
B		yes / no
C		yes / no
D		yes / no
E		yes / no
F		yes / no

4b. For each use, name the swatch letter you would pick and give two pieces of evidence from your own test data.

The use	My swatch	Evidence 1 (from my tests)	Evidence 2 (from my tests)
A winter hat			
A soccer jersey			
A dress shirt for a job interview			
A swimsuit			

Exit card (tear off or write on an index card)

3 things you learned about a fabric with your hands today:

1. _____ 2. _____ 3. _____

2 fibers you can now tell apart, and how:



1 question about fibers or care:



Teacher key

Part 1a. Natural: cotton, wool, silk, linen. Manufactured: polyester, nylon, rayon, spandex. Rayon is the one students miss; it starts as wood pulp, which is a plant, but it is processed into fiber in a factory, so it is counted as manufactured.

Part 1b. Cotton: a plant, the fluff around the seed of the cotton plant. Wool: an animal, the fleece of a sheep (other animal fibers include alpaca, mohair from goats, and cashmere). Silk: an insect, the thread a silkworm spins for its cocoon. Linen: a plant, the stem of the flax plant.

Part 1c, expected answers.

Fiber	Warmth	Water	Wrinkles	Stretch	Cost and care
Cotton	Some	Absorbs a lot, dries slowly	Wrinkles	None on its own	Cheap; can shrink; machine wash
Wool	High, even when damp	Absorbs, dries slowly	Springs back, resists wrinkles	Some, springy	Costly; hand wash or dry clean; hot water felts it
Silk	Some	Absorbs; water can leave a mark	Wrinkles some	Little	Costly; hand wash or dry clean
Linen	Low, coolest in heat	Absorbs a lot, dries fast	Wrinkles the most	None	Costly; gets softer with age
Polyester	Some, traps heat	Beads or wicks, does not absorb, dries fast	Resists wrinkles	Little on its own	Cheap; machine wash; melts with heat; holds odor
Nylon	Some	Beads, dries fast	Resists wrinkles	Some	Cheap; strong; melts with heat
Rayon	Low	Absorbs, weak when wet	Wrinkles	Little	Mid cost; often hand wash
Spandex	Low	Beads	Resists wrinkles	High, springs back	Always blended; dryer heat



					destroys it
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Part 2, the station set. Label the swatches A to F in this order or your own; keep the answer envelope matched to the labels.

Swatch	Fiber	What students should find
A	Cotton	Smooth to soft, cool, thin; no stretch; water soaks straight in
B	Wool	Rough or fuzzy, warm, thick; springs back a little; water beads at first then soaks slowly
C	Silk or linen	Silk: very smooth, cool, thin. Linen: slightly rough, cool, thicker slubs; both absorb
D	Polyester (fleece or shirting)	Smooth or fuzzy, warm, varies; little stretch; water beads and sits on top
E	Nylon (windbreaker or tights)	Slick, cool, thin; some stretch; water beads and runs off
F	Spandex blend (leggings)	Smooth, warm, thick; stretches a lot and comes back; water beads

Score the station record complete, partial, or missing: complete means three tests filled for at least four swatches, with a real use named (not a color) for each.

Part 3, the burn test results. These are the standard identification results.

Fiber	Flame	Smell	Residue
Cotton	Burns fast, keeps burning	Burning paper or leaves	Soft gray ash that crumbles
Wool	Burns slowly, curls away, often self-extinguishes	Burning hair	Black bead that crushes to powder
Polyester	Does not burn, melts and shrinks away, drips	Sweet and chemical	Hard bead that will not crush

The melt sentence: because melted fiber sticks to skin and keeps burning it, so the burn is worse and it cannot be brushed off.

Part 4b, defensible answers. The fabric named matters less than the evidence.

Use	Best choice and why
Winter hat	Wool (B). Warm even when damp, springs back, holds its shape. A fleece polyester (D) is also defensible: warm, cheap, dries fast
Soccer jersey	Polyester (D) or nylon (E). Water beads and moves away instead of soaking in, dries fast, light. Cotton is the wrong answer and the tests show why: it soaked the drop right in and stays wet
Dress shirt for a job interview	Cotton (A) for breathability and the look, or a cotton and polyester blend for wrinkle resistance. The reason has to come from the tests
Swimsuit	Spandex blend (F). Stretches and comes back, water beads, dries fast, and it has to move with a body

Standards checked by this handout: NYS ATPD 1. Selection a) and 3. Maintenance a); CTE HSW 4. Classroom and Workplace Safety a); CTE SUS 2. Environmental Management e); NYS FACS Standard 2 (Intermediate) PI C; CDOS 3a Basic Skills; National FCS 3.0 16.2.1, 16.2.2, and 16.2.4.